Code-Layout, Readability and Reusability

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates OptiCrop's codebase in terms of structure, readability, and reusability.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Follows standard PEP 8, 4-space indentation
2	Proper File Structure	Files and folders are logically organized	Yes	Separate data/, model/, templates/, static/, notebooks/ folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., FEATURES, X_train, scaler, best_model
4	Function / Method Names	Functions are descriptively named	Yes	e.g., predict_crop(), home(), find_your_crop()
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings and inline comments in train_model.py and app.py
6	Modular Design	Code is split into reusable functions/modules	Yes	Dataset generation, training, and app logic kept in separate files
7	No Redundant Code	Duplicate or unused code is removed	Partial	Notebook intentionally mirrors train_model.py for exploratory/teaching purposes
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic try/except around prediction route; could add stricter input validation

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	train_model.py	Python / scikit-learn	Can retrain the model on any updated crop_data.csv	High
2	predict_crop() function	Python / Pickle	Reusable for testing new inputs without running the Flask app	Medium
3	style.css	CSS	Shared across Home, About, and Find Your Crop templates	High
4	scaler.pkl	scikit-learn StandardScaler	Reused on every prediction call without refitting	High
5	generate_dataset.py	Python / NumPy / Pandas	Can regenerate or extend the synthetic dataset with new crops/ranges	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of data, model, and app logic
Readability	4	Descriptive names and comments throughout
Reusability	4	Training script and model artifacts are easily reused independently
Documentation / Comments	4	Docstrings present; app.py could use a few more inline explanations
Overall Score	4	Solid, maintainable structure for a project of this scope